

According to IPsoft, Amelia has the ability to function at one-fourth the billing rates for human engineers and could challenge even the most innovative of human resources departments while handling tasks.

Here comes a 3D-printed robot

At the inaugural Code Conference, organised at Rancho Palos Verdes, Calif, a cute, 60cm tall figure, called Jimmy, walked on to the stage, introduced itself and waved its arms. Jimmy is a 3D-printed robot from the house of Intel.

Although Jimmy is just a research robot so far, the company has plans to sell the fully-customisable, 3D-printable robot kit in the market by the year end. The consumer version of Jimmy will be priced starting around US\$ 1600 (about ₹ 95,000) and will be sold via 21stcenturyrobot.com. The kit will have all the parts that cannot be 3D printed, including the motors, batteries, processor and wires. The hardware design will be freely available so anyone with a 3D printer can print it.

The consumer model of Jimmy robot runs on Intel Edison platform, which is a low-cost computer on a chip, much like Arduino but with much wider functionality. Jimmy can perform a range of tasks including singing, dancing, tweeting, translating and, most importantly, delivering cold beers. It is an open source robot and allows the developers to create their own apps and users to download them



so that they can run whatever they want on their machines. Intel will also make a marketplace for such apps and designs.

Here's a video (please scan the QR code) showing how Jimmy works!

New 2D transistors pave way for faster electronics

Researchers have now come up with world's first two-dimensional field-effect transistor (FET) which will enable the development of faster electronics.

These 2D FETs do not have the issue of performance drop-off under high voltages and can provide high electron mobility, even if they are scaled to a monolayer in thickness.

Fabrication of the 2D heterostructures was done by the layers of a transition metal dichalcogenide, hexagonal boron nitride and graphene arranged up by relatively weak electric forces or van der Waals that attract neutral molecules to one another in gases.

The bonding of the interfaces via van der Waals and the usage of a multi-step transfer process prepares a platform for making complex devices based on crystalline layers. It removes the constraints of lattice parameters that tend to restrict the performance and growth of conventional heterojunction materials. The study was published in journal *ACS Nano*.

A new wonder material to replace graphene

Graphene is undoubtedly no less than a wonder material that could transform our lives in the days to come. However, cadmium arsenide is even better. Here's how...

Graphene has a major drawback that, it is functional only in two dimensions. This means building complex pieces of hardware out of the material is not such an easy task. Taking note, researchers from Oxford, Stanford and the Lawrence Berkeley National Laboratory decided to continue looking and found cadmium arsenide, an unlikely saviour that could very well prove to be a viable alternative to graphene.

For starters, it can transmit electricity at tremendous speeds, much like graphene. However, unlike the latter, cadmium arsenide can also work in all three dimensions. This means building transistors and sensors out of this material in all probability will be much



Artist's view of one of the Google's 180 satellites (Courtesy: www.rt.com)

easier. Additionally, cadmium arsenide could be a good candidate for everyday use, claim the researchers.

Graphene is increasingly being used in a plethora of applications in electrical industry. It could be used to develop better batteries as also in creating super-strong yet flexible screens. However, major drawbacks do deter its use on a large scale.

Google's 180 satellites to bring Internet access across the world

Looks like Google's fascination with bringing the Internet access to inhospitable areas around the world knows no bounds. A recent report by the *Wall Street Journal* has revealed that Google is looking to build a total of 180 small, high-capacity satellites and set them out into low orbit so as to aid in providing the Internet connections.

These satellites could reportedly entail a cost of a \$1-\$3 billion. But once in orbit these are more likely to pay for themselves. Further, if all goes well, Google could double its satellite count, so that more and more areas could be covered. This would be an extension to Google's Project Loon through which it aims to substantially increase its ad revenue by bringing more and more people online.

Google earlier acquired Titan Aerospace, a company that makes solar-powered drones, which was in acquisition talks with Facebook just a few months ago. Google is eyeing to use the technology provided by Titan to assist its Project Loon, which aims to connect people to the Internet in far flung areas.

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